

11. (1.75 points)

An actuary is using a GLM bootstrap model to estimate the reserves for a line of business. The model residuals by accident year and development period are:

	Model Residuals as of (months)				
Accident Year	12	24	36	48	60
2013	100	-100	-50	0	50
2014	100	-100	-20	10	
2015	-80	80	0		
2016	45	-70			
2017	0				

a. (0.5 point)

Briefly describe heteroscedasticity and identify an issue it may cause when using a GLM bootstrap model.

b. (0.75 point)

Evaluate whether development period heteroscedasticity exists.

c. (0.5 point)

Describe a method to adjust for heteroscedasticity.

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SPRING 2018 EXAM 7, QUESTION 11

TOTAL POINT VALUE: 1.75

LEARNING OBJECTIVE(S): A7-A9

SAMPLE ANSWERS

Part a: 0.5 point

Sample 1

Definition: residuals do not share a common variance.

Issue: GLM assume residuals are iid (*independent identically distributed*). If not, we cannot sample the residuals from any part of the triangle.

Sample 2

Heteroscedasticity means the variance of residuals are not the same for the triangle

It violates the main assumption that residuals are independent and identically distributed.

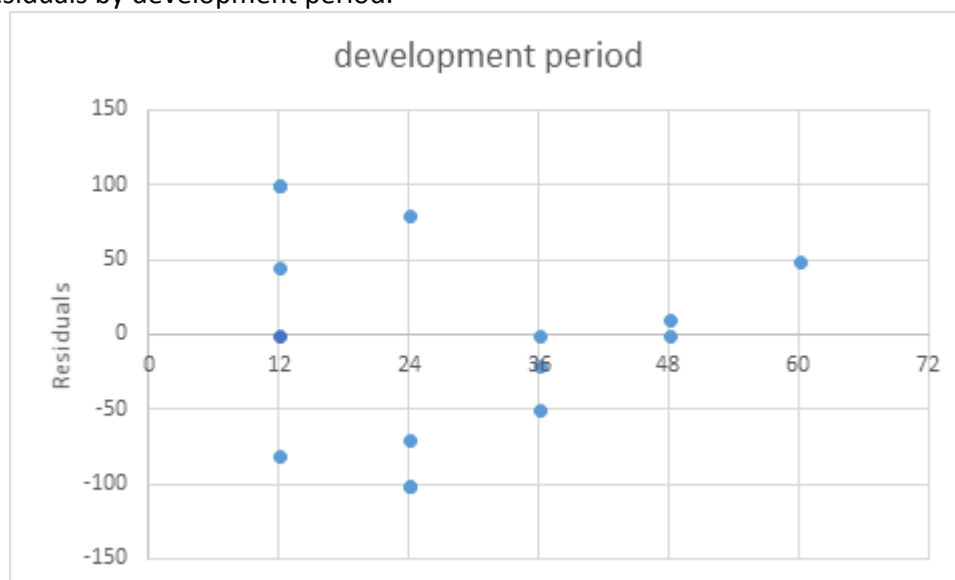
Sample 3

Residuals are assumed to be independently and identically distributed. Therefore, variance of residuals should be constant. Heteroscedasticity is when variance of residuals it not constant. If the variance is not constant, some projected losses will be overestimated and some underestimated.

Part b: 0.75 point

Sample 1

Plot the residuals by development period.



From the plot above, heteroscedasticity exists since development periods 36-60 have a smaller spread/variance than 12-24.

Sample 2

Mean of residuals @ 12mo = $(100 + 100 + -80 + 45 + 0) / 5 = 33$

Mean of residuals @ 24mo = -47.5

Mean of residuals @ 36mo = -23.33

Mean of residuals @ 48mo = 5

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Variance of residuals @ 12mo = $(100^2 + 100^2 + -80^2 + 45^2 + 0^2) / 5 - 33^2 = 4,596$

Variance of residuals @ 24mo = 5,569

Variance of residuals @ 36mo = 422

Variance of residuals @ 48mo = 25

stdev @ 12mo = $\sqrt{4,596} = 67.8$

stdev @ 24mo = 74.6

stdev @ 36mo = 20.5

stdev @ 48mo = 5

There appears to be development period heteroscedasticity as the standard deviation is significantly different between 12-24 and 36-48.

Sample 3

Heteroscedasticity seems to exist for development periods. Residuals in age 12 and 24 range from 100 to -80 and -100 to 80 respectively whereas the residuals in age 36 and 48 is more narrowly ranged.

Part c: 0.5 point

Sample 1

Stratified Sampling: Group residuals with similar variance and sample residuals from each group separately.

Sample 2

Using hetero-adjustment.

- i. Calculate the hetero adjustment factor after grouping residuals.
 $h_i = \text{Total standard deviation} / \text{standard deviation of group } i$.
- ii. Multiply h_i to standardized residuals in group i .
- iii. Sample the adjusted residuals as usual.
- iv. Un-adjust the hetero adjustment after sampling.

Sample 3

Use scale parameters.

Determine groups that show similar variance. Calculate the scale parameter to adjust the residuals. The scale parameter is $\sqrt{\theta_{\text{total}}} / \sqrt{\theta_i}$ where $\theta_i = \sum [\sqrt{N/(N-P)} \times r]^2 / n_i$. This would give hetero-adjusted residuals. After sampling, apply the reverse to get a residual for the development period.

EXAMINER'S REPORT

Candidates were expected to understand what heteroscedasticity is, the issues associated with it, methods to identify the existence of it, and techniques to adjust residuals when it does exist.

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Part a

Candidates were expected to know the definition of heteroscedasticity as well as issues associated with it. For the latter, multiple responses were accepted as long as the candidate showed understanding of the issue and it was tied to the GLM bootstrap model.

Common mistakes included:

- Stating "It is not homoscedastic" without showing further or sufficient understanding of the GLM bootstrap model framework.
- Not mentioning residuals when defining heteroscedasticity in the GLM bootstrap model framework (although credit was given if the candidate did not specifically mention residuals but did imply it in other parts of the question).
- Stating that the incremental losses are iid instead of residuals being iid.
- Talking about non-constant distribution instead of non-constant variance. Distributions can be different but still have the same variance.

Part b

Candidates were expected to know methods to identify the existence of heteroscedasticity. Candidates needed to state that the variance/spread/range of the later development periods were smaller than the early periods and provide proof backing up the argument.

Common mistakes included:

- Only stating that variance is different without providing support.
- Evaluating for heteroscedasticity along accident year or calendar year, since the question specifically asked for development year.
- Forgetting to tie the change in variance of residuals back to whether or not heteroscedasticity exists.
- Stating that there are patterns or trends in the residuals. Heteroscedasticity requires non-constant variances. If patterns or trends exist, that does not necessarily imply that the variance is changing.
- Including/using the sum of squared residuals. Although similar to variance, they are not the same thing.

Part c

Candidates were expected to know techniques to adjust residuals when heteroscedasticity does exist. The candidate was only expected to know the steps and theory behind these techniques, not the actual formulas and equations.

Common mistakes included:

- Only stating "split into groups" or "group like residuals" when talking about stratified sampling. Grouping development periods into groups with similar variances was required for full credit. If the candidate had already implied groups with similar variances in part b, full credit was given for part c.
- Not removing the adjustment factor after sampling. If the adjustment factor is not removed, the sampled residuals are biased.